GhostHire — Project Documentation

Evidence-Based Opportunity Verification & Threat Intelligence Platform

Hack-to-ween 2026 • Open Innovation & Real-World Solutions • AWS Serverless

1. Executive Summary & Problem Statement

Across college campuses worldwide, millions of students and fresh graduates actively seeking internships and entry-level jobs fall prey to predatory recruitment scams. Scammers exploit placement anxiety by sending unsolicited WhatsApp messages, Telegram alerts, and fake PDF appointment letters impersonating reputable companies (Google, Microsoft, Nexora Labs).

The Critical Market Gaps:

- **The 'Dark Social' Blindspot:** Scams circulate as chat messages and screenshot images, completely bypassing corporate email firewalls and traditional anti-phishing gateways.
- **Opaque AI Chatbots:** Generic LLM chatbots frequently hallucinate or provide vague, untraceable advice rather than deterministic, auditable evidence.
- **Information Isolation:** Fraudsters spam multiple college batches simultaneously while isolated students evaluate offers without cross-campus peer warnings.

2. The GhostHire Solution Ecosystem

GhostHire delivers a tripartite defense mechanism combining **serverless OCR**, **explainable heuristic scoring**, and **crowdsourced community intelligence**:

Module	Technology	Core Responsibility
WhatsApp Screenshot Scanner	Amazon Textract & Clipboard API	Allows drag-and-drop or Ctrl+V image paste. Extracts chat bubbles, recruiter handles, and links without manual typing.
Explainable Risk Engine	Deterministic Heuristic Rulebook	Calculates a 0–100 risk score and quotes exact suspicious phrases (e.g. 'Pay ■999 fees', 'within 30 mins').
Community Pattern Radar	Amazon DynamoDB NoSQL	Decentralized crowdsourced registry where students check domains and submit consensual fraud alerts in sub-10ms.
1-Click WhatsApp Shield	Native Clipboard Formatter	Generates formatted warning summaries for students to forward into batch WhatsApp groups to stop viral fraud.

3. Explainable Risk Engine & Heuristic Weight Matrix

GhostHire calculates risk deterministically using a bounded sum of weighted heuristic detectors plus community penalties:

Mathematical Model:

$$\textbf{Risk Score} = \min\left(100, \sum (w_i \times I(\text{Signal}_i)) + \text{Community_Penalty}\right)$$

Where w_i is the heuristic weight and $I(\text{Signal}_i) \in \{0, 1\}$ indicates detector activation.

Signal Identifier	Points	Detection Logic & Regex Triggers	Student Guidance

UPFRONT_PAYMENT	**+25 pts**	Registration, training, kit fees, security deposits, UPI payment demands (■500 - ■10,000).	Legitimate employers pay you; they NEVER charge fees. Reject offer.
DOMAIN_MISMATCH	**+20 pts**	Recruiter claiming top brand using free webmail (@gmail, @yahoo) or lookalike domain (*-careers.xyz).	Real corporate recruiters exclusively use official enterprise domain emails.
GUARANTEED_SELECTION	**+15 pts**	Claims of 100% selection or direct hire bypassing technical evaluation rounds.	Reputable companies evaluate problem-solving skills; guaranteed hiring is a classic trap.
SUSPICIOUS_URL	**+15 pts**	Obfuscated or shortened links (bit.ly, tinyurl, .xyz, .top, .tk).	Shortened links conceal destination phishing sites. Navigate only via official portals.
CREDENTIAL_REQUEST	**+15 pts**	Demands for Aadhaar OTP, net banking passwords, or identity documents before offer contract.	High risk of financial identity theft. Never share OTPs or passwords.
ARTIFICIAL_URGENCY	**+10 pts**	Extreme pressure timers ('Pay within 30 minutes', 'Offer expires in 1 hour').	Scammers induce panic to bypass rational reflection. Pause and verify.

4. AWS Serverless Cloud Architecture

GhostHire is engineered on a 100% serverless, cloud-native AWS infrastructure designed for zero-idle cost, high scalability, and tamper-evident auditability:

AWS Service	Resource Config	Operational Architecture & Workflow
Amazon Textract	DetectDocumentText API	Processes uploaded WhatsApp chat images, PDF appointment letters, and clipboard pastes. Automatically identifies text lines, phone numbers, and URLs with layout preservation.
Amazon S3	Bucket: ghost.hire Region: ap-southeast-2	Serves as the Evidence Vault. Stores encrypted screenshot images with SHA-256 integrity hashes and generates 5-minute presigned upload/download URLs for zero backend proxy overhead.
Amazon DynamoDB	Table: GhostHireReports Region: ap-southeast-2	Serverless NoSQL database backing the Community Pattern Radar. Stores verified scam reports and lookalike recruiter domains with sub-10ms query latency and automatic scaling.
AWS Lambda & Mangum	Python 3.11 Runtime / ASGI	Packages the FastAPI application into an event-driven serverless Lambda handler, ensuring automatic horizontal scaling and zero cost during idle periods.

5. RESTful API Endpoint Reference

All endpoints are prefixed with /api and support CORS for seamless web integration:

Endpoint	Method	Payload / Parameters	Response & Behavior
/api/analyze	POST	{"text": string}	Runs explainable heuristic engine. Returns risk_score, risk_level, recommendation, and itemized findings.
/api/analyze/upload	POST	multipart/form-data (file)	Uploads screenshot to Amazon S3, executes Amazon Textract OCR, and evaluates extracted text against risk detectors.
/api/community/{domain}	GET	Path: domain (string)	Queries Amazon DynamoDB for verified community fraud reports and notes associated with the target domain.
/api/report	POST	{"domain", "notes", "user_consent"}	Records anonymized scam intelligence report into DynamoDB for cross-campus early warning detection.
/api/config	GET	None	Returns active AWS connectivity flags, bucket name, region, and serverless feature health status.

6. Innovation & Competitive Differentiation

GhostHire directly addresses the limitations of both traditional enterprise security tools and generic generative AI chatbots:

Capability	Traditional Spam Filters	Generic AI Chatbots	GhostHire Platform
WhatsApp Screenshot OCR	No (Email only)	Partial (Token costly)	**Yes (Amazon Textract)**
Explainable Score & Evidence	Binary (Spam / Ham)	Hallucination prone	**Deterministic 0–100**

Community Scam Radar	None	None (Static knowledge)	**Yes (AWS DynamoDB)**
1-Click WhatsApp Shield	None	None	**1-Tap Group Alert**
Cost per Verification	High Enterprise License	High LLM Token Costs	**< $0.001 / scan**

7. Test Verification & Quality Assurance

GhostHire maintains a comprehensive automated test suite with 100% pass rates across unit and integration suites:

- **Test Suite Execution:** 8/8 tests passing (test_risk_engine.py & test_api_integration.py).
- **Live AWS Verification:** S3 bucket (ghost.hire) and DynamoDB table (GhostHireReports) confirmed operational in ap-southeast-2.
- **Security Hardening:** Zero credentials committed to version control; .gitignore enforced across all credential files.